

LEVEL 3 — ADVANCED BUSINESS DATA ANALYTICS SYSTEM

Cognevance Technologies

Data Analysis with Python — Project Report

1. Project Overview

The Advanced Business Data Analytics System project focuses on analyzing a large-scale retail transaction dataset to understand business performance, revenue trends, customer behavior, key performance indicators, and future revenue patterns.

The project was developed using Python, with Pandas and NumPy for data preprocessing and numerical analysis, Matplotlib for data visualization, and Scikit-learn for predictive analytics and revenue forecasting.

The complete workflow includes large-scale data ingestion, data cleaning, preprocessing, feature engineering, exploratory data analysis, KPI analysis, trend analysis, customer behavior analysis, predictive modeling, model evaluation, predictive insights, and business recommendations.

This project follows the **Level 3 — Advanced Business Data Analytics System** requirements provided by Cognevance Technologies.

2. Project Objectives

The main objectives of this project are:

- To collect and analyze a large-scale retail business dataset.
- To perform advanced data preprocessing and data validation.
- To perform feature engineering using transaction and date information.
- To use Pandas, NumPy, Matplotlib, and Scikit-learn.
- To analyze revenue trends and business KPIs.
- To analyze customer purchasing behavior.
- To identify high-value and high-frequency customers.
- To develop a predictive analytics model for revenue forecasting.
- To evaluate the predictive model using appropriate metrics.
- To generate predictive insights and business recommendations.
- To document the complete analytics workflow and architecture.

These objectives correspond to the Level 3 project requirements.

3. Dataset Overview

The project uses the **Online Retail II** dataset from the **UCI Machine Learning Repository**.

Attribute	Details
Dataset Name	Online Retail II
Dataset Domain	Retail / E-Commerce
Original Records	1,067,371
Original Columns	8
Final Records After Cleaning	779,425
Final Analytical Columns	18
Programming Language	Python
Environment	Google Colab
Main Libraries	Pandas, NumPy, Matplotlib, Scikit-learn

Dataset Columns

The original dataset contains the following fields:

- Invoice
- StockCode
- Description
- Quantity
- InvoiceDate
- Price
- Customer ID
- Country

These fields provide information about invoices, products, quantities, transaction dates, prices, customers, and countries.

4. Data Loading and Ingestion

The dataset was provided in an Excel workbook containing two sheets:

- Year 2009-2010
- Year 2010-2011

Both sheets were loaded using Pandas and combined into a single DataFrame for analysis.

The original combined dataset contained:

1,067,371 records and 8 columns.

The data types and dataset structure were inspected before performing preprocessing and analysis.

5. Data Cleaning and Preprocessing

The dataset was cleaned and validated using Pandas.

The following data-quality operations were performed.

5.1 Duplicate Records

The dataset initially contained:

34,335 duplicate records.

These duplicate rows were removed before further analysis.

5.2 Missing Values

Missing values were identified in:

- Description
- Customer ID

Rows containing missing Description or Customer ID values were removed because these fields were required for the planned product and customer analysis.

After cleaning, no missing values remained.

5.3 Invalid Transaction Values

The following fields were checked for invalid values:

- Quantity
- Price

The dataset contained:

18,390 records with non-positive Quantity values.

70 records with non-positive Price values.

These invalid records were removed.

5.4 Final Data Quality

After preprocessing:

- Final records: **779,425**
- Final columns: **8 original columns before feature engineering**
- Missing values: **0**
- Duplicate rows: **0**
- Invalid Quantity values: **0**
- Invalid Price values: **0**

The cleaned dataset was then used for feature engineering and analysis.

6. Feature Engineering

Feature engineering was performed to create additional variables required for business analysis.

6.1 Revenue Calculation

Revenue was calculated using:

Revenue = Quantity × Price

This created a new **Revenue** column.

6.2 Date-Based Features

The InvoiceDate field was used to create:

- Year
- Month
- Month_Name

- Quarter
- Day
- DayOfWeek
- Day_Name
- Hour
- Is_Weekend

After feature engineering, the analytical dataset contained:

779,425 records and 18 columns.

These features were used for revenue trends, KPI analysis, customer analysis, and predictive analytics.

7. Technology and Library Usage

The following Python libraries were used:

Library	Purpose
Pandas	Data loading, cleaning, transformation, grouping and analysis
NumPy	Numerical calculations and statistical operations
Matplotlib	Data visualization and trend analysis
Scikit-learn	Train/test splitting and predictive modeling
Google Colab	Development and execution environment

The installed versions used during analysis were:

- Pandas — 2.2.3
 - NumPy — 2.1.3
 - Matplotlib — 3.10.0
 - Scikit-learn — 1.6.1
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8. Exploratory Data Analysis

Exploratory analysis was performed to understand overall business performance and identify important patterns.

The analysis included:

- Total revenue analysis
 - Quantity analysis
 - Transaction analysis
 - Customer analysis
 - Country-wise revenue analysis
 - Year-wise revenue trends
 - Monthly revenue trends
 - Customer transaction frequency
 - Customer revenue contribution
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9. KPI Analysis

The following key performance indicators were calculated.

KPI	Value
Total Revenue	17,374,804.27
Total Quantity Sold	10,513,952
Total Transactions	36,969
Total Customers	5,878

These KPIs provide an overall summary of the business performance represented in the dataset.

10. Revenue Trend Analysis

Revenue was analyzed across years and months to identify business trends.

10.1 Year-wise Revenue

Year	Revenue
2009	683,504.01

2010 8,374,496.0
9

2011 8,316,804.1
6

The analysis shows a substantial increase in revenue from 2009 to 2010, followed by a similar revenue level in 2011.

10.2 Monthly Revenue Analysis

Monthly revenue was aggregated across all available years.

The highest aggregate monthly revenue was:

November — 2,322,665.63

The lowest aggregate monthly revenue was:

February — 950,643.88

This analysis indicates noticeable differences in sales performance across different months.

11. Customer Behavior Analysis

Customer behavior was analyzed using transaction frequency and customer revenue contribution.

11.1 Most Frequent Customers

The top customer by number of transactions was:

Customer ID 14911 — 398 transactions

Other high-frequency customers included:

- Customer 12748 — 336 transactions
- Customer 17841 — 211 transactions
- Customer 15311 — 208 transactions
- Customer 13089 — 203 transactions

This analysis helps identify customers with high purchasing frequency.

11.2 Highest Revenue Customers

The highest-revenue customer was:

Customer ID 18102 — 580,987.04

Other high-revenue customers included:

- Customer 14646 — 528,602.52
- Customer 14156 — 313,437.62
- Customer 14911 — 291,420.81
- Customer 17450 — 244,784.25

This analysis identifies customers making significant contributions to overall revenue.

12. Country-wise Business Analysis

Revenue was also analyzed by country.

The highest-revenue country in the dataset was:

United Kingdom — 14,389,234.92

Other high-revenue countries included:

- EIRE
- Netherlands
- Germany
- France
- Australia
- Spain
- Switzerland
- Sweden
- Denmark

The analysis shows that the United Kingdom represents the largest revenue contribution among the countries analyzed.

13. Predictive Analytics

A predictive analytics model was developed to forecast monthly revenue.

A **Linear Regression** model from Scikit-learn was used.

The historical monthly revenue data was divided chronologically into:

- Training data — first 20 months
- Testing data — final 5 months

The forecasting features used were:

- Year
- Month

The model was trained using historical revenue values and then used to predict revenue for the final five months.

14. Predictive Model Results

The actual and predicted revenue values were compared.

Month	Actual Revenue	Predicted Revenue
August 2011	644,051.04	711,077.43
September 2011	950,690.20	740,059.68
October 2011	1,035,642.45	769,041.92
November 2011	1,156,205.61	798,024.17
December 2011	517,208.44	827,006.41

The model evaluation results were:

R² Score: -0.1830

RMSE: 262,374.67

The average prediction error across the five test months was:

29.84%

The results indicate that the Linear Regression model provides a baseline forecasting approach but has limited predictive accuracy for this dataset.

15. Predictive Insights

The predictive analysis produced the following insights:

1. The highest actual revenue during the test period was recorded in **November 2011**, with revenue of **1,156,205.61**.
 2. The highest predicted revenue during the test period was **827,006.41**, predicted for December 2011.
 3. The model underestimated several high-revenue months, particularly September, October, and November 2011.
 4. The model overestimated December 2011 revenue compared with the actual value.
 5. The average prediction error of **29.84%** indicates that the current model has limitations and could be improved using more advanced forecasting techniques.
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16. Business Recommendations

Based on the analysis, the following recommendations can be made:

16.1 Focus on High-Revenue Periods

Historical revenue trends should be monitored to identify strong sales periods and support business planning.

16.2 Prepare for Peak Demand

Inventory and marketing activities can be planned before historically strong revenue periods to support potential increases in demand.

16.3 Retain High-Frequency Customers

Customers with high transaction frequency, such as Customer 14911, should receive appropriate retention and engagement strategies.

16.4 Focus on High-Value Customers

High-revenue customers such as Customer 18102 can be prioritized for customer retention and relationship-building activities.

16.5 Monitor Business KPIs

Revenue, quantity sold, transactions, and customer counts should be monitored regularly to identify changes in business performance.

16.6 Improve Forecasting Accuracy

The current Linear Regression model can be treated as a baseline. More suitable time-series forecasting techniques can be considered to improve future prediction accuracy.

17. Data Visualizations

The project includes visualizations created using Matplotlib.

The major visualizations include:

17.1 Year-wise Revenue Trend

A line chart showing revenue performance across 2009, 2010, and 2011.

17.2 Monthly Revenue Trend

A line chart showing aggregate revenue across months.

17.3 KPI Analysis

A visualization showing the major business KPIs.

17.4 Top Customer Analysis

A bar chart showing customers with the highest revenue contribution.

17.5 Top Customer Transaction Analysis

A bar chart showing customers with the highest transaction frequency.

17.6 Country-wise Revenue Analysis

A bar chart showing the highest-revenue countries.

17.7 Actual vs Predicted Revenue

A visualization comparing actual revenue with predicted revenue for the test period.

These visualizations were created using Python and Matplotlib.

18. Complete Analytics Workflow and Architecture

The complete analytics workflow followed in this project was:

Large-Scale Retail Dataset

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Data Collection

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Data Ingestion

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Data Cleaning & Preprocessing

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Missing Value & Duplicate Handling

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Data Validation

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Feature Engineering

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Exploratory Data Analysis

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KPI & Trend Analysis

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Customer Behavior Analysis

↓

Country-wise Analysis

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Predictive Analytics

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Model Evaluation

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Predictive Insights

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Business Recommendations

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Dashboard & Visualization

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Final Documentation & Report

This workflow represents the complete analytics architecture implemented for the project.

19. Project Architecture

The system follows a structured analytics architecture:

Data Layer

Online Retail II Dataset

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Processing Layer

Pandas + NumPy

Data cleaning, validation, transformation, and feature engineering.

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Analytics Layer

Pandas + NumPy

KPI analysis, trend analysis, customer analysis, and country analysis.

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Visualization Layer

Matplotlib

Charts, revenue trends, KPI visualizations, and analytical outputs.

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Machine Learning Layer

Scikit-learn

Train/test splitting, Linear Regression, prediction, and model evaluation.

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Insight Layer

Predictive insights and business recommendations.

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Reporting Layer

Dashboard outputs, documentation, and final project report.

20. Project Deliverables

The completed project contains:

- Complete Python analysis notebook
- Large-scale retail dataset
- Data cleaning and preprocessing code
- Feature engineering code
- Exploratory data analysis
- KPI analysis
- Customer behavior analysis
- Revenue trend analysis
- Predictive analytics model
- Model evaluation results
- Matplotlib visualizations
- Predictive insights
- Business recommendations
- Project documentation

- Final project report
- GitHub repository

These deliverables correspond to the expected outputs specified for the **Level 3 — Advanced Business Data Analytics System** project.

21. Conclusion

The **Advanced Business Data Analytics System** project successfully demonstrates an end-to-end business analytics workflow using Python.

A large-scale retail transaction dataset containing more than one million original records was loaded, cleaned, validated, and transformed for analysis. Feature engineering was performed to generate revenue and date-based analytical features.

The project analyzed important business KPIs, revenue trends, customer purchasing behavior, customer revenue contribution, and country-wise performance.

A Linear Regression model was also developed for monthly revenue forecasting and evaluated using R^2 and RMSE. The model produced an R^2 score of **-0.1830**, an RMSE of **262,374.67**, and an average prediction error of **29.84%**, indicating that the model can serve as a baseline but requires improvement for more accurate forecasting.

The project generated practical business insights related to high-revenue periods, high-frequency customers, high-value customers, KPI monitoring, and forecasting improvement.

Overall, the project fulfills the requirements of **Level 3 — Advanced Business Data Analytics System** and demonstrates practical skills in:

Data Preprocessing → Feature Engineering → Exploratory Data Analysis → KPI Analysis → Customer Analysis → Visualization → Predictive Modeling → Model Evaluation → Predictive Insights → Business Recommendations → Analytics Documentation

Project Information

Attribute	Details
Project	Advanced Business Data Analytics System

Level	Level 3 — Advanced
Organization	Cognevance Technologies
Domain	Retail / E-Commerce Analytics
Language	Python
Environment	Google Colab
Libraries	Pandas, NumPy, Matplotlib, Scikit-learn
Dataset	Online Retail II
Original Records	1,067,371
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Original Columns	8
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Predictive Model	Linear Regression
R ² Score	-0.1830
RMSE	262,374.67
Average Prediction Error	29.84%